Electrical & Automation Master Plan: Phased Build-Out

Project Location: 270 Bolla Ave

Status: Finalized Directive for Rough-In

Note on Scope: This document is intended to define the High-Voltage (HV) infrastructure, panel distribution, and structural rough-in requirements. It is not an exhaustive list of KNX or DALI-2 programming or device-level requirements.

1. Project Vision & Phasing Strategy

The home is transitioning from a standard grid-tied residence to a software-defined, battery-first environment. We are executing this in stages to ensure the home is operational now while making future high-power upgrades effortless.

- **Phase 1 (Current):** Install 320A service, three SPAN Smart Panels, and the DALI-2/KNX Load Control infrastructure. We will have minimal wall switches (only what is strictly required by code) to maintain a clean aesthetic while sensor technology is finalized.
- **Phase 2 & 3 (Future):** Integration of a high-capacity Hybrid Inverter, a dedicated "Battery Wall," and Solar PV arrays.

The Goal: Phase 1 wiring must ensure this equipment can be dropped in without tearing open walls or resizing conduits.

2. Garage Infrastructure (The "Energy Hub")

Strategic use of garage wall space is critical. We are planning a dedicated **"Energy Wall"** (approximately 48" x 60") for the future inverter and solar equipment, plus a separate **"Battery Wall"** area.

Wall Prep & Finishing

- **Reinforced Mounting Surface:** Fully insulate, drywall, and paint the North wall of the garage.
- **Plywood Backing:** Before drywall installation, install 3/4-inch CDX Plywood backing directly to the studs.
- **Rationale:** This allows for secure mounting of the heavy Inverter (~120 lbs) and future Battery stacks anywhere on the wall without being limited by stud locations, ensuring a professional, finished look.

Conduit & Wiring Prep

- **Conduit Upsizing:** Use 2.5-inch primary conduits between the panels and the future inverter location to allow for effortless pulling of heavy-gauge wire later.
- **Sub-Panel Feeder:** For the 100A feed to the North Wall (Sub-SPAN), use #1 AWG Aluminum (SER).
- **Main Service Entry:** The 320A (400A Class) service feeds the main panels. In Phase 2, this will pass through the Hybrid Inverter as the primary power mixer.

3. Panel Distribution & Labeling

We are using three SPAN Smart Panels (96 total circuits) to ensure total telemetry.

Panel	Location	Primary Loads
Main SPAN #1	Garage	EV Chargers (60A), Induction Range (50A), Ovens (50A), HVAC.
Main SPAN #2	Garage	Kitchen, Laundry, **LCP-1 Feeds (inc. 3x Spares)**.
Sub-SPAN	North Wall	Left Wing Rooms, **LCP-2 Feeds (inc. 3x Spares)**, Pergola (18kW).

Labeling Standard: All circuits in all panels must be labeled using **printed magnetic tape labels**. Hand-written ink or pen on the panel covers is prohibited.

4. The "LCP Rule": Automated vs. Manual

To maximize data density, loads are separated based on how they are switched.

Automated via LCP-1/LCP-2 (Example Loads)

Generally, any load that glows, moves, or heats (and lacks internal logic) is a candidate for LCP integration.

- **Required Loads:** All lighting/accent LEDs, QuietCool Fans, motorized skylights, and towel warmers.
- **Attic Service Loops (Future Proofing):** Pull **three spare NM-B-PCS Duo (14/2 + DALI pair)** runs from each LCP (**6 total**) into the attic space.
    - **Execution:** Coil 10–15 feet of slack in a central, accessible attic location.
    - **Termination:** Per code, these must be terminated in a labeled 4-inch square junction box with a cover. Do not leave "loose" ends outside of a box.
- **Circuit Standard:** Use 15 Amp circuits for LCP automation feeds. This facilitates the use of Southwire NM-B-PCS Duo.

Bypass LCP (Direct to SPAN)

Only "always-on" loads or those with internal controls:

- Receptacles.
- HVAC.
- Major Appliances (Fridge, Ovens, Range, etc.).

5. KNX Low-Voltage Infrastructure (30V Bus)

The control layer for the home utilizes a 30V KNX bus. To support a "minimalist switch" aesthetic today while allowing for full manual control tomorrow, we are implementing a "Wall-Tap" service loop strategy.

- **Universal Bus Routing:** The KNX Green Bus wire must be routed through **every wall** in every living area at a standard switch height (typically 48" AFF).
- **Wall Service Loops:** At the center of every wall segment (or designated entry points), provide a 24-inch service loop coiled loosely behind the drywall.
- **Tapping Logic:** Loops must not be pulled tight. They should have enough slack to be fished or pulled to a different location on that same wall segment if a switch is added later.
- **Documentation:** The exact location of these loops must be photographed and measured from a fixed point (e.g., corner or door frame) before drywall installation to ensure they can be located via magnet or measurement.

6. Life Safety & Future Additions

- **Fire/CO Detection:** A separate, dedicated 15A circuit for Nest Protect units (Smoke/CO/CO_2) throughout the house.
- **Pergola Heaters (18kW):** Three 30A 240V circuits landed on the Sub-SPAN. These are monitored for grid-blending but are not routed through the lighting LCPs. A standalone DALI-2 control pair will be pulled from LCP-2 to trigger the contactors.
- **Trash Compactor:** Removed from scope. (Circuit #13 from original Right Wing notes is deleted).

7. Technical Standards for the Team

- **Strict Load Separation:** Receptacles and Lighting must never share a circuit.
- **No Switch Legs:** Do not pull traditional switch-leg wires to every wall. Install only the minimum code-required switches.
- **Box Depth:** Standardize on 4-inch square, 2-1/8" deep steel boxes for all junctions to accommodate DALI relay pucks.
- **Conduit Hub:** Ensure all 2.5-inch conduits terminate in a large junction box or "gutter" near the future inverter location for easy Phase 2 integration.